

Feature-based 2.5D Controlled Source Electromagnetic Inversion using Generative Priors

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Abstract—In this study, we investigate feature-based 2.5D controlled source electromagnetic (CSEM) inversion using generative priors. Two-point-five dimensional modeling using finite difference method (FDM) is adopted to compute the response of horizontal electric dipole (HED) excitation in the 2D survey. Generative priors are integrated into the inversion process by reparameterizing the model with a variational autoencoder (VAE). During the inversion, the model is iteratively updated using the Gauss-Newton method and projected onto the VAE decoder's range. Both numerical and field experiments demonstrate that the proposed inversion approach effectively incorporates prior knowledge, enhances reconstruction accuracy, and exhibits strong generalization capabilities.

Index Terms—2.5D electromagnetic modeling, 2.5D electromagnetic inversion, controlled source electromagnetics, generative models, deep learning, generative priors, constrained inversion, joint inversion

I. INTRODUCTION

Electromagnetic (EM) methods have been widely used in geophysical prospecting. Among them, the controlled source electromagnetic (CSEM) method has been successfully used to evaluate the hydrocarbon reservoirs [1]–[14]. This method involves stimulating the low-frequency EM by electric dipoles, measuring the EM at selected locations, and inverting and interpreting the EM measurements with reasonable data inversion methods.

Accurate CSEM modeling is essential for reliable data inversion and interpretation. Due to the lack of translational symmetry in the horizontal electric dipole (HED) source, the EM field in CSEM surveys cannot be accurately represented in a purely 2D space. A direct solution is to perform full 3D modeling [3], [9], but this approach is computationally expensive and resource-intensive. Alternatively, if the geological structure is assumed to vary only in two dimensions—remaining constant along one direction—2.5D modeling provides an efficient compromise [2], [6], [7]. It enables accurate representation of the EM field while significantly reducing computational cost, making it suitable for practical inversion and interpretation tasks.

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EM inversion is inherently ill-posed and nonlinear, often leading to non-unique or unstable solutions. To mitigate this, prior knowledge is typically incorporated to constrain the inversion. A common strategy is to add regularization terms, such as the L_1 norm or Tikhonov regularization, [15]–[17]. However, simple regularizations may struggle to capture more complex prior information. An alternative approach involves reparameterizing the model in a compact transformed space, which can be constructed using various techniques. For instance, model-based inversion uses horizons and polygons to define regions with assumed uniform properties. The unknowns include the physical parameters within each region and the coordinates of boundary control points [18]–[20], which are typically optimized using gradient-based methods. While this reduces the number of unknowns, it is often sensitive to the initial model and prone to local minima.

Another strategy involves reparameterizing the model with a linear combination of global basis functions, such as Fourier or wavelets. This method has been applied in EM inversion [21]–[24] and seismic tomography [25]–[28]. However, selecting appropriate basis functions is often heuristic and lacks the flexibility to encode task-specific prior knowledge. The basis functions can also be constructed with machine learning methods. In hydrogeologic ERT inversion, for example, complex model patterns are described using a training dataset, and singular value decomposition (SVD) is applied to learn representative image bases [29].

Deep learning provide a powerful framework to distill the accumulation of human experience and historical data into neural models and conduct effective inference on similar tasks [9], [28]. Generative modeling [30] allows for learning the underlying data distribution and generating new samples consistent with prior knowledge. Unlike supervised learning, generative models are typically trained solely in the model space, independent of physical simulations. As a result, paired data and model samples are not required for constructing the training set. In recent years, generative models have been explored for inversion tasks across various domains, including biomedical imaging [31], [32], electromagnetic inverse scattering [33], [34], image processing [35], [36], and geophysical applications [37], [38]. Among the commonly used generative models, variational autoencoders (VAEs) and generative adversarial networks (GANs) learn nonlinear mappings from the original data space to a lower-dimensional latent space, providing a compact representation that serves as a generative prior in the inversion process [30].

The generative prior can be integrated into the inversion

process by directly cascading the generative model with the physical forward operator in the objective function [33], [34], [38], [39]. In this framework, gradients are directly back-propagated from the objective function to the latent variables. However, the nonlinearity of the generative model can lead the optimization process to become trapped in local minima [40]. This challenge becomes even more pronounced when the physical forward operator is also nonlinear, further complicating the optimization landscape. To address this, some methods (e.g., projected gradient descent PGD) iteratively project the model updated via gradient-based optimization back onto the prior space, typically defined by the range of a VAE decoder or GAN generator [32], [36], [40]–[42]. However, these methods are primarily designed for linear inverse problems and rely on first-order optimization. When the forward operator is nonlinear and the solution space is 2.5D or 3D, the computational cost of first-order methods may become prohibitive.

In this work, we investigate a feature-based 2.5D CSEM inversion regularized by a generative prior. A 2.5D simulation is employed to model the EM field generated by a horizontal electric dipole (HED). A variational autoencoder (VAE) is trained on a generated dataset to capture prior knowledge of the model space. To perform the inversion, a projected Gauss-Newton (PGN) method is used to optimize the conductivity model, ensuring consistency with both the observed data and the learned prior. Numerical and field experiments demonstrate the effectiveness of the proposed approach.

II. METHOD

A. 2.5D forward problem and jacobian computation

Starting from the frequency domain Maxwell's equations, and neglecting the magnetic source, and using the j -notation, we have

$$\nabla \times \bar{\mathbf{E}} = -j\omega\mu\bar{\mathbf{H}}, \quad (1a)$$

$$\nabla \times \bar{\mathbf{H}} = j\omega\epsilon'\bar{\mathbf{E}} + \bar{\mathbf{J}}, \quad (1b)$$

where $\bar{\mathbf{E}}$, $\bar{\mathbf{H}}$ and $\bar{\mathbf{J}}$ are respectively the electric field, magnetic field and electric source in the spatial domain. ω is the angular frequency, μ is the permeability, and $\epsilon' = \epsilon - j\frac{\sigma}{\omega}$ is the complex permittivity in which σ is the conductivity.

In a typical 2.5D survey, the EM field has to be modelled in the 3D space (e.g., the response of the dipole excitation), while the distribution of physical parameter is only variant in the 2D plane but invariant along the third dimension. Without loss of generality, we assume the model extends along the x direction, and the EM field in the yo z plane is of interest. The solving of 2.5D problem generally involves transforming the field and source with Fourier transform along the spatial invariant dimension (x -direction), that is,

$$\hat{\mathbf{F}}(k_x, y, z) = \int_{-\infty}^{+\infty} \bar{\mathbf{F}}(x, y, z) e^{-jk_x x} dx, \quad (2a)$$

$$\bar{\mathbf{F}}(x, y, z) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{\mathbf{F}}(k_x, y, z) e^{jk_x x} d(k_x), \quad (2b)$$

where $\hat{\mathbf{F}}$ denotes the field or source in the wave-number domain, and $\mathbf{F}$ can be either $\mathbf{E}$, $\mathbf{H}$ or $\mathbf{J}$. Rewriting (1a)

and (1b) in the wave-number domain, and neglecting the z component of $\mathbf{J}$, we have

$$\frac{\partial \hat{E}_z}{\partial y} - \frac{\partial \hat{E}_y}{\partial z} = -j\omega\mu\hat{H}_x, \quad (3a)$$

$$\frac{\partial \hat{E}_x}{\partial z} - jk_x \hat{E}_z = -j\omega\mu\hat{H}_y, \quad (3b)$$

$$jk_x \hat{E}_y - \frac{\partial \hat{E}_x}{\partial y} = -j\omega\mu\hat{H}_z, \quad (3c)$$

$$\frac{\partial \hat{H}_z}{\partial y} - \frac{\partial \hat{H}_y}{\partial z} = (j\omega\epsilon + \sigma)\hat{E}_x + \hat{J}_x, \quad (3d)$$

$$\frac{\partial \hat{H}_x}{\partial z} - jk_x \hat{H}_z = (j\omega\epsilon + \sigma)\hat{E}_y + \hat{J}_y, \quad (3e)$$

$$jk_x \hat{H}_y - \frac{\partial \hat{H}_x}{\partial y} = (j\omega\epsilon + \sigma)\hat{E}_z. \quad (3f)$$

Further organizing (3a)-(3f), we have

$$\begin{aligned} & -\frac{\partial}{\partial y} \left(\frac{y_0}{k_e^2} \frac{\partial \hat{E}_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{y_0}{k_e^2} \frac{\partial \hat{E}_x}{\partial z} \right) + jk_x \frac{\partial}{\partial y} \left(\frac{1}{k_e^2} \right) \frac{\partial \hat{H}_x}{\partial z} \\ & - jk_x \frac{\partial}{\partial z} \left(\frac{1}{k_e^2} \right) \frac{\partial \hat{H}_x}{\partial y} + y_0 \hat{E}_x = -\hat{J}_x + \frac{\partial}{\partial y} \left(\frac{jk_x}{k_e^2} \hat{J}_y \right) + \frac{\partial}{\partial z} \left(\frac{jk_x}{k_e^2} \hat{J}_z \right), \end{aligned} \quad (4a)$$

$$\begin{aligned} & -\frac{\partial}{\partial y} \left(\frac{z_0}{k_e^2} \frac{\partial \hat{H}_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{z_0}{k_e^2} \frac{\partial \hat{H}_x}{\partial z} \right) + jk_x \frac{\partial}{\partial z} \left(\frac{1}{k_e^2} \right) \frac{\partial \hat{E}_x}{\partial y} \\ & - jk_x \frac{\partial}{\partial y} \left(\frac{1}{k_e^2} \right) \frac{\partial \hat{E}_x}{\partial z} + z_0 \hat{H}_x = \frac{\partial}{\partial y} \left(\frac{z_0}{k_e^2} \hat{J}_z \right) - \frac{\partial}{\partial z} \left(\frac{z_0}{k_e^2} \hat{J}_y \right), \end{aligned} \quad (4b)$$

where $k_e^2 = k_x^2 - k^2$, $k^2 = -z_0 y_0$, $z_0 = j\omega\mu$, $y_0 = j\omega\epsilon + \sigma$. Discretizing (4a) and (4b) with finite difference method (FDM), a set of linear equations can be derived and solved with respect to the x component electric and magnetic fields in the wave-number domain ($\hat{E}_x$ and $\hat{H}_x$). Further resorting to (3a)-(3f), the y and z components of $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ can be computed, and the spatial domain fields $\bar{\mathbf{E}}$ and $\bar{\mathbf{H}}$ can be finally computed using the inverse Fourier transform as shown in (2b).

In this work, we focus on a marine CSEM survey, in which inline electric measurements ($\bar{E}_y$) are collected along the single survey line (extending along the y direction) by a ship. Both the transmitters and receivers are HEDs, whose orientations are along the survey line direction. We use the function $\mathcal{F}(\cdot)$ to represent the relationship between the conductivity σ and the measured $\bar{E}_y$ at the receivers.

The Jacobian matrix can be computed with adjoint method [2], [43]. The relationship between the adjoint field, adjoint source and the sensitivity can be formulated with [43]

$$\int_D (\bar{\mathbf{J}}_m^+ \cdot \frac{\partial \bar{\mathbf{H}}}{\partial \sigma_k} + \bar{\mathbf{J}}_e^+ \cdot \frac{\partial \bar{\mathbf{E}}}{\partial \sigma_k}) d\nu = \int_D \bar{\mathbf{E}}^+ \cdot \bar{\mathbf{E}} \phi_k d\nu, \quad (5)$$

where ϕ_k is the basis function at the k -th pixel, D is the effective domain of ϕ_k , and $\bar{\mathbf{J}}_m^+$, $\bar{\mathbf{J}}_e^+$ and $\bar{\mathbf{E}}^+$ are respectively adjoint magnetic source, electric source and electric field. The sensitivity of $\bar{E}_y$ to conductivity σ can be written with

$$\frac{\partial \bar{E}_y}{\partial \sigma} = \int_{-\infty}^{+\infty} \int_D \bar{\mathbf{E}}_{J_y}^+ \cdot \bar{\mathbf{E}} ds dy, \quad (6)$$

where $\bar{\mathbf{E}}_{J_y}^+$ is the adjoint electric field excited with y -direction electric source. Transforming both $\bar{\mathbf{E}}_{J_y}^+$ and $\bar{\mathbf{E}}$ to wave-number domain with (2a) and after some derivation, (6) can be expressed with

$$\frac{\partial \bar{E}_y}{\partial \sigma} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_D \hat{\mathbf{E}}_{J_y}^+(-k_x, y, z) \cdot \hat{\mathbf{E}}(k_x, y, z) ds dk_x. \quad (7)$$

Noticing that $\bar{E}_x$ is an odd function of k_x while $\bar{E}_y$ and $\bar{E}_z$ are even functions of k_x , (7) can finally expressed with

$$\frac{\partial \bar{E}_y}{\partial \sigma} = \frac{1}{\pi} \int_0^{+\infty} \int_D (-\hat{E}_{J_y,x}^+ \hat{E}_x + \hat{E}_{J_y,y}^+ \hat{E}_y + \hat{E}_{J_y,z}^+ \hat{E}_z) ds dk_x. \quad (8)$$

B. Variational autoencoder (VAE)

The variational autoencoder (VAE) is a typical generative neural network structure, which is able to learn the distribution from the training dataset and generate new realizations conforming to the prior distribution [44], [45]. Generally, the VAE structure is composed of an encoder $\mathcal{E}$ and a decoder $\mathcal{D}$. The encoder input is the data sampled from the prior distribution, and the encoder outputs respectively describe the mean and variance of a high-dimensional Gaussian distribution. Denoting the encoder input with $\mathbf{m}$, the output can be expressed with

$$\mathcal{E}(\mathbf{m}) = (\mathcal{E}_1(\mathbf{m}), \mathcal{E}_2(\mathbf{m})) = (\boldsymbol{\mu}_{\mathcal{E}}, \log \boldsymbol{\sigma}_{\mathcal{E}}^2), \quad (9)$$

where $\boldsymbol{\mu}_{\mathcal{E}}$ and $\boldsymbol{\sigma}_{\mathcal{E}}^2$ are respectively the mean and variance of a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_{\mathcal{E}}, \boldsymbol{\sigma}_{\mathcal{E}}^2)$, from which the latent variable $\mathbf{v}$ is sampled:

$$\mathbf{v} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathcal{E}}, \boldsymbol{\sigma}_{\mathcal{E}}^2). \quad (10)$$

$\mathbf{v}$ is then input into the decoder $\mathcal{D}$, yielding the reconstruction of $\mathbf{m}$:

$$\tilde{\mathbf{m}} = \mathcal{D}(\mathbf{v}). \quad (11)$$

The training objectives of the VAE are two-fold: making the reconstruction loss of VAE as small as possible, and simultaneously make the latent space approximate a standard normal distribution as close as possible. The objective function in VAE training can be expressed with

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \sum_{q=1}^Q (\boldsymbol{\mu}_{\mathcal{E},q}^2 + \boldsymbol{\sigma}_{\mathcal{E},q}^2 - \log \boldsymbol{\sigma}_{\mathcal{E},q}^2 - 1). \quad (12)$$

Here, $\mathcal{L}_{\text{rec}}$ is the reconstruction loss, Q is the dimension of the latent space, and λ is the coefficient adjusting the relative weight of the reconstruction loss and Kullback-Leibler divergence loss. For a more detailed derivation of the objective function please refer to [44]. After the training is finished, the decoder $\mathcal{D}$ can be regarded as a data generator that characterizes the prior distribution and is able to generate new realizations conforming to the prior distribution.

C. Feature-based 2.5D CSEM inversion

In the deterministic EM inversion framework, the 2.5D CSEM inversion can be formulated as optimizing the unknown model $\mathbf{m}$ by minimizing a loss function composed of both the data misfit term and the Tikhonov regularization [46], i.e.,

$$\mathcal{L}(\mathbf{m}) = \|\mathbf{W}_d(\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathbf{m}))\|^2 + \alpha \|\mathbf{W}_m \mathbf{m}\|^2, \quad (13)$$

and

$$\hat{\mathbf{m}} = \arg \min_{\mathbf{m} \in \mathcal{S}} \mathcal{L}(\mathbf{m}). \quad (14)$$

Here, $\mathbf{m}$ is the 2D conductivity distribution in the seabed, $\mathbf{d}_{\text{obs}}$ is the measured complex value y component electric field as introduced in Section II-A, and $\mathcal{S} \subseteq \mathbb{R}^{N_m}$ is a set that captures the data structures that $\mathbf{m}$ is *a priori* required to obey (e.g., non-negativity). In this work, we adopt a more informative generative prior described by the range of VAE decoder ($\mathcal{R}(\mathcal{D})$), i.e.,

$$\hat{\mathbf{m}} = \arg \min_{\mathbf{m} \in \mathcal{R}(\mathcal{D})} \mathcal{L}(\mathbf{m}). \quad (15)$$

As suggested by existing works [33], [34], [38], [39], one possible approach to solve (15) is to reformulate the optimization problem in the latent space $\mathbb{R}^Q$, i.e.,

$$\hat{\mathbf{v}} = \arg \min_{\mathbf{v} \in \mathbb{R}^Q} \|\mathbf{W}_d(\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathcal{D}(\mathbf{v})))\|^2 + \alpha \|\mathbf{W}_m \mathcal{D}(\mathbf{v})\|^2, \quad (16)$$

and obtain the optimized model $\hat{\mathbf{m}}$ from the optimized latent variable $\hat{\mathbf{v}}$, i.e.,

$$\hat{\mathbf{m}} = \mathcal{D}(\hat{\mathbf{v}}). \quad (17)$$

However, we find (from both reports in [40], [41] and our numerical experiments) that (16) can be easily stuck in local minima, since the cascade of two nonlinear operators exaggerate the ill-posedness of the problem. References [40], [41] solve (15) with projected gradient-descent (PGD), that is, in each iteration $\mathbf{m}$ is first updated with gradient descent method and then projected to the $\mathcal{R}(\mathcal{D})$ set. However, first-order optimizations can take hundreds or even thousands of iterations to converge. Since in a 2.5D EM survey, each forward modeling and gradient calculation requires solving dozens of linear systems with tens of thousands of unknowns, higher-order optimizations (e.g., Gauss-Newton) are highly preferred to accelerate the convergence.

We solve (16) with a projected Gauss-Newton (PGN) method [47], [48]. Similar to PGD, the PGN can also be separated into two stages. At the k -th iteration, $\mathbf{m}$ is first updated with Gauss-Newton method, that is

$$\mathbf{m}_{k+1} = \mathbf{m}_k - \mathbf{H}_k^{-1} \mathbf{g}_k, \quad (18)$$

where $\mathbf{H}_k$ and $\mathbf{g}_k$ are respectively Hessian matrix and gradient of $\mathcal{L}(\mathbf{m})$ with respect to $\mathbf{m}$ at the k -th iteration:

$$\mathbf{H}_k = \mathbf{J}_k^T \mathbf{W}_d^T \mathbf{W}_d \mathbf{J}_k + \alpha \mathbf{W}_m^T \mathbf{W}_m, \quad (19a)$$

$$\mathbf{g}_k = -\mathbf{J}_k^T \mathbf{W}_d^T \mathbf{W}_d (\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathbf{m}_k)) + \alpha \mathbf{W}_m^T \mathbf{W}_m \mathbf{m}_k, \quad (19b)$$

where $\mathbf{J}_k$ is the Jacobian matrix calculated by (5)-(8). In the second stage, the updated $\mathbf{m}_k$ is projected to $\mathcal{R}(\mathcal{D})$ by solving the following minimization problem:

$$\hat{\mathbf{v}}_{k+1} = \arg \min_{\mathbf{v}} \|\mathbf{m}_{k+1} - \mathcal{D}(\mathbf{v})\|_2^2, \quad (20)$$

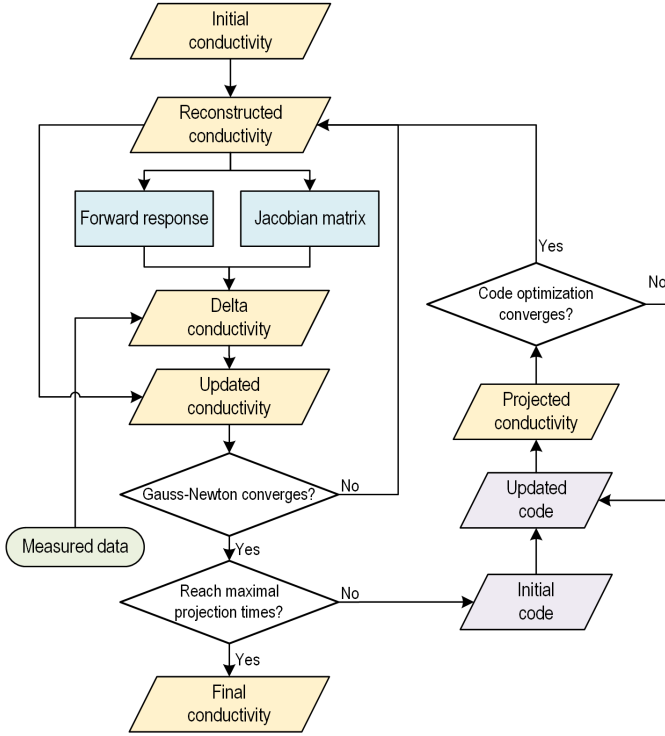


Fig. 1: The workflow of the proposed inversion.

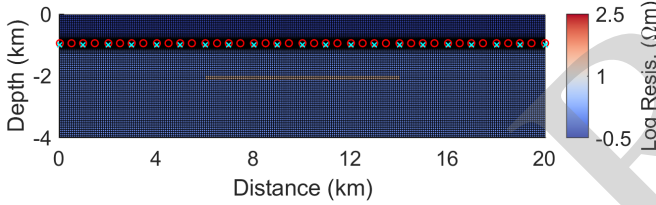


Fig. 2: The configuration of grid, transmitter and receivers plotted.

and compute the $\hat{\mathbf{m}}_{k+1}$ with (17). The inversion terminates when the data misfit reaches a predefined level or exceeds the maximum iterations. The workflow of the proposed inversion is shown in Fig. 1. We would like to point out that in the inversion, the projection is only conducted at several *projection steps* (the total number is generally smaller than 5 and with intervals between each other) instead of each iteration. The aim is to minimize the generalization risk of VAE therefore reducing the possibility that the inversion is trapped in the local minima. In addition, because the total number of projections is relatively small, and the time to solve (20) is not a computational bottleneck compared to the time for forward modeling and gradient computation, we directly use first-order gradient descent (e.g., adaptive moment estimation (ADAM, [49])) to solve (20).

III. NUMERICAL EXPERIMENTS

A. Implementations

We setup a domain of investigation (DoI) of 20 km wide and 4 km deep. The shallow part is the 1 km sea water and the deep part is the 3 km seabed. Forty-one inline electric

dipoles are uniformly distributed at the depth of 950 m, excited one at a time, and twenty-one line receivers are uniformly distributed on the sea floor. We adopt a non-uniform mesh, which is densified at the area of transmitter-receiver and uniform elsewhere, with 251 grids in the horizontal direction and 156 grids in the vertical direction. The schematic of the mesh, as well as the distribution of transmitters and receivers are shown in Fig. 2. The transmitters are denoted with red circles, and the receivers are denoted with cyan crosses. The adopted frequencies are 0.25 Hz and 0.75 Hz, and noise is added to the synthetic data with (21),

$$d_{\text{obs},mnk}^{\text{noise}} = d_{\text{obs},mnk}(1 + \beta(u_1 + iu_2)), \quad (21)$$

where β is the noise level, and u_1 and u_2 are independently sampled from uniform distribution $\mathcal{U}(-1, 1)$. m, n and k are respectively the index of transmitter, receiver and frequency.

The adopted VAE in this work is fine-tuned from the k1 – 8f VAE introduced in the noted latent diffusion work [50], and the structure schematic is illustrated in Fig. 3. The VAE have 86 million trainable parameters and 17 million non-trainable parameters in total. The three channel input and output are respectively merged into one channel by averaging parameter values along the channel direction. A fine-tune dataset containing 50000 images is simulated, 90% used for training and 10% for validation. Each image has a size of 128×128 , and has a pattern of one or two thin resistive layers exist in the uniform conductive background. This corresponds to the *a priori* knowledge that, the DoI contains a thin and layer-shaped resistive hydrocarbon reservoir residing in the uniform conductive seabed, but the location, shape and size of the layer is to be determined in the inversion. The attribute value is set to 0 for the background and sampled from uniform distribution $\mathcal{U}(1.3, 2)$ for the layers. Some examples in the training dataset is shown in Fig. 4. The reconstruction loss $\mathcal{L}_{\text{rec}}$ as included in (12) is composed of the L_2 loss both in the pixel domain and feature domain, i.e.,

$$\mathcal{L}_{\text{rec}} = \|\mathbf{m} - \tilde{\mathbf{m}}\|_2^2 + \gamma \|\mathcal{G}(\mathbf{m}) - \mathcal{G}(\tilde{\mathbf{m}})\|_2^2, \quad (22)$$

where $\mathcal{G}(\cdot)$ denotes a pre-trained and frozen image processing neural network such as AlexNet [51], and γ is the hyper-parameter adjusting the relative weights of the two types of loss. The inclusion of the feature domain loss aims to better capture the detailed features of small-sized layers. The fine-tuning converges in about 10 steps, and the performance of the fine-tuned VAE is tested. First, the ground truth models $\mathbf{m}_{\text{rec}} = \mathcal{D}(\mathcal{E}_1(\mathbf{m}))$ (see Fig. 4). Besides, the continuity of the latent space is also checked: given two models $\mathbf{m}_1$ and $\mathbf{m}_2$ and their corresponding latent variables $\mathbf{v}_1 = \mathcal{E}_1(\mathbf{m}_1)$ and $\mathbf{v}_2 = \mathcal{E}_1(\mathbf{m}_2)$, the intermediate models $\mathbf{m}_{1,2,p} = (1 - p)\mathcal{D}(\mathbf{v}_1) + p\mathcal{D}(\mathbf{v}_2)$ are visualized in Fig. 5. Both the observations that the reconstructions in Fig. 4 well resemble the ground truths, and that the reconstructed models can transform smoothly between each other in Fig. 5, verify the efficacy of the VAE.

We introduce some metrics to measure the inversion quality, including data misfit (DM), intersection over union (IoU),

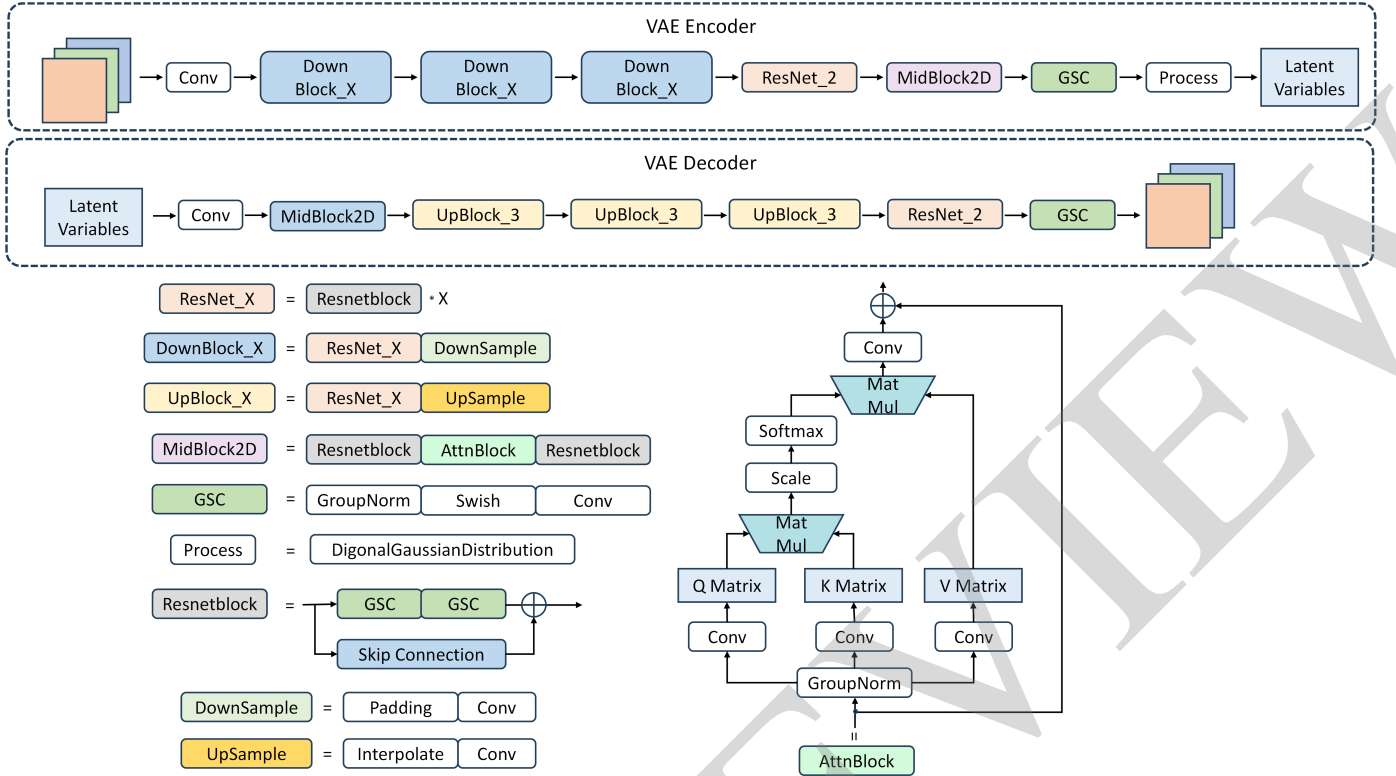


Fig. 3: The structure of the VAE.

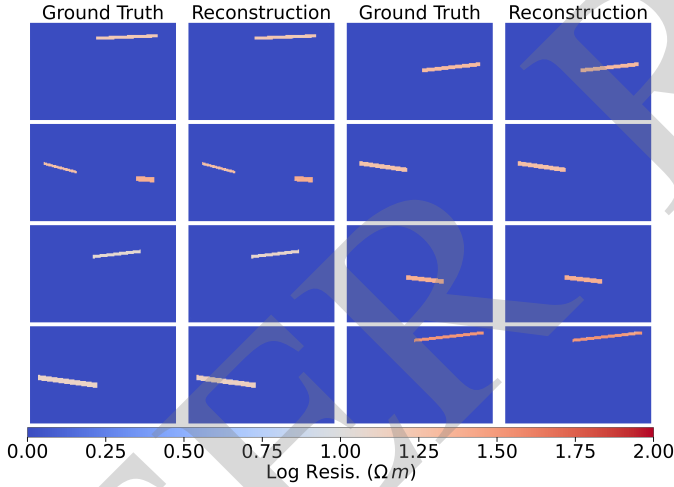


Fig. 4: Visualization of some example images in the training dataset and corresponding reconstructions.

mean square error (MSE) and structural similarity (SSIM) [52], as listed in (23)-(26):

$$DM = \frac{\|\mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{rec}}\|_2}{\|\mathbf{d}_{\text{obs}}\|_2}, \quad (23)$$

and

$$\text{IoU}(\mathbf{I}_{\text{rec}}, \mathbf{I}_{\text{gnd}}) = \frac{|\mathbf{I}_{\text{rec}} \cap \mathbf{I}_{\text{gnd}}|}{|\mathbf{I}_{\text{rec}} \cup \mathbf{I}_{\text{gnd}}|}, \quad (24)$$

$$\text{MSE} = \|\mathbf{m}_{\text{gnd}} - \mathbf{m}_{\text{rec}}\|_2^2, \quad (25)$$

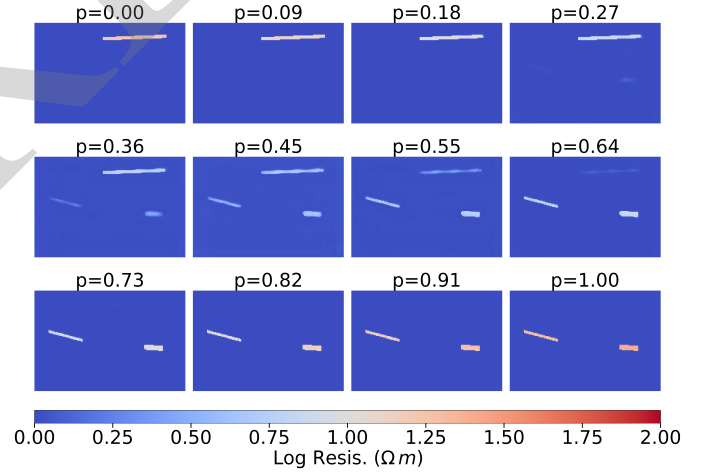


Fig. 5: Visualization of intermediate models transferring between two ground truth models in Fig. 4.

$$\text{SSIM} = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \quad (26)$$

Here, $\mathbf{d}_{\text{obs}}$ and $\mathbf{d}_{\text{rec}}$ are respectively observed and reconstructed data. $\mathbf{I}_{\text{gnd}}$ is the 0-1 identification function, in which 1 pixels indicate the target reservoir and 0 pixels indicate the background in the ground truth model. $\mathbf{I}_{\text{rec}}$ is also a 0-1 identification function, in which 1 pixels indicate where the amplitude of the reconstructed model is greater than half the peak value. $\mathbf{m}_{\text{gnd}}$ and $\mathbf{m}_{\text{rec}}$ are respectively the ground truth model and reconstructed model. $\mu_x, \mu_y, \sigma_x, \sigma_y$ and σ_{xy} denote the mean, standard deviation and covariance between

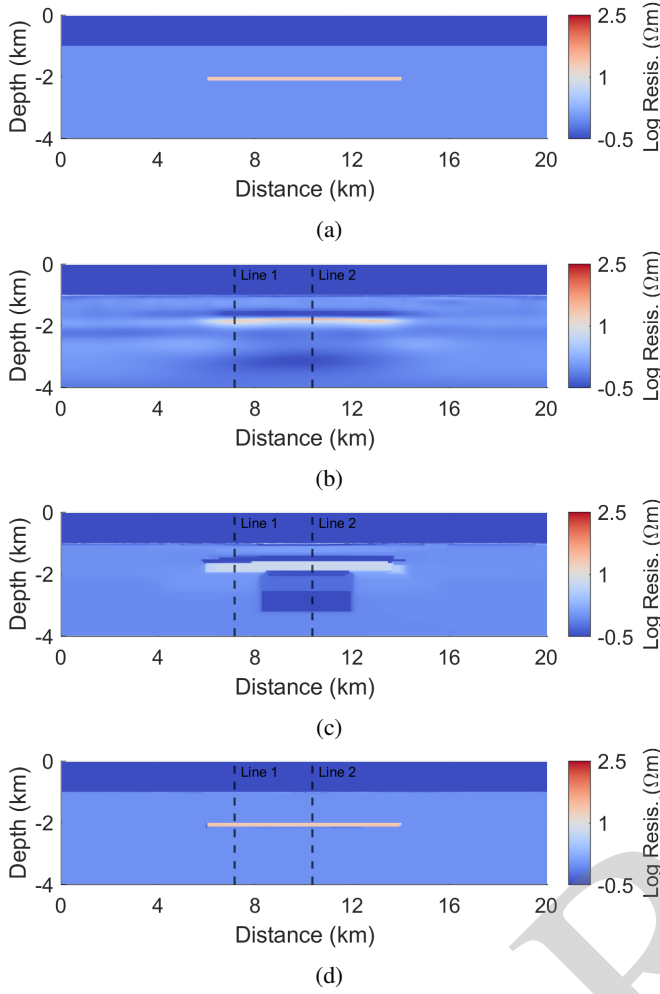


Fig. 6: 6a: the ground truth model, 6b: the L_2 inversion model, 6c: the WL_2 inversion model, 6d: feature-based inversion.

two images x and y , and C_1 and C_2 are two positive constants.

B. Experiment 1

In experiment 1, the ground truth model is shown in Fig. 6a. The configurations of the DoI, mesh and transmitters and receivers are introduced in Section III-A. A long thin resistive layer is located 1 km below the seabed and has a length of 8 km, and the thickness is 120 m. The conductivities of sea water, seabed and resistive layer are respectively 3 S/m, 1 S/m and 0.05 S/m, respectively. The initial seabed has a uniform conductivity of 0.8 S/m. We respectively conducted inversions with conventional L_2 regularization, weighted L_2 regularization and the studied feature-based inversion (abbreviated L_2 , WL_2 and FB from this point onwards). The L_2 and WL_2 reconstructions are respectively shown in Fig. 6b and 6c. For a clearer comparison, we display the resistivity distributions of different reconstructions at two traces (whose locations are shown by black dash lines in Fig. 6a) in Fig. 7a and 7b. In both L_2 and WL_2 reconstructions, the resistive layer can be clearly identified with distinguishable boundaries, and the location and size are basically accurate. However, in the conventional inversions, the resistive layer is reconstructed

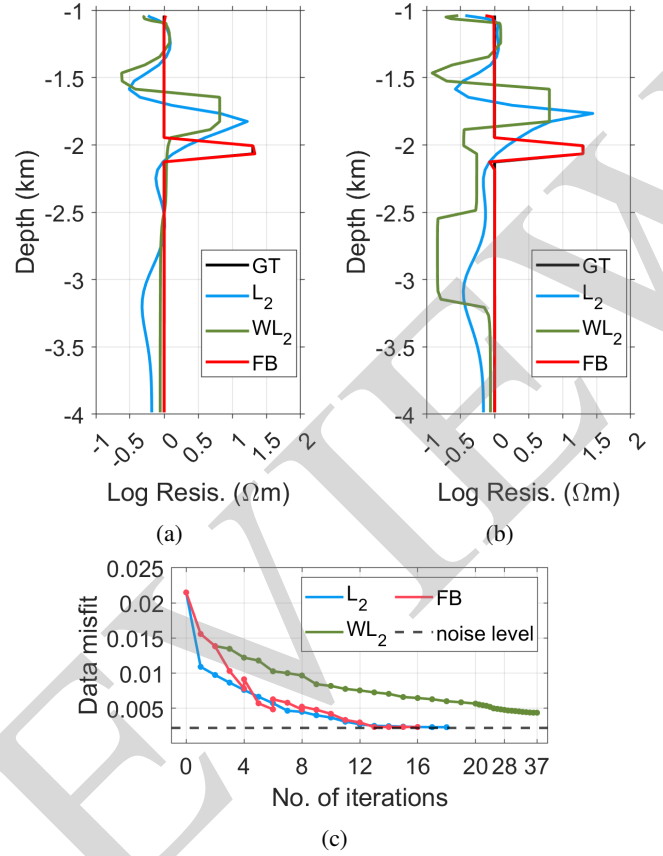


Fig. 7: Fig. 7a and 7b show the resistivity curves reconstructed by different methods at the two traces (the location are denoted with black dash lines in Fig. 6a), and Fig. 7c shows the curves of data misfit in the inversions.

shallower than their actual depths, with a deviation of about 300 to 400 m, and the layer thickness is exaggerated. For example, in WL_2 reconstruction the resistive layer is about twice the real thickness. Besides, conductive anomalies can be observed in both Fig. 6b and 6c, one of them is close to the upper surface of the resistive layer, and the other is below the resistive layer with offset ranging from 8 km to 12 km and depth at about 3 km. We point out that similar artifact patterns in 2.5D mCSEM inversion can also be observed in previous works (for instance, see Figure 7 of [2]), which hinders the accurate evaluation of the reservoir and clearly shows the non-uniqueness of the solution.

Fig. 6d displays the FB reconstruction, and evident improvement can be observed compared to the conventional inversions, e.g., in Fig. 7a and 7b, the ground truth curves (plotted in black) are almost totally overlapped by the FB reconstruction (plotted in red). The different types of artifacts as shown in conventional inversions are eliminated to a great margin, providing a much unique background and distinguishable target whose depth, thickness and attribute value highly resemble the ground truth. The convergence curves of the data misfit are shown in Fig. 7c. Both L_2 and FB inversions converge in about 15 steps with the final data misfit close to the noise level,

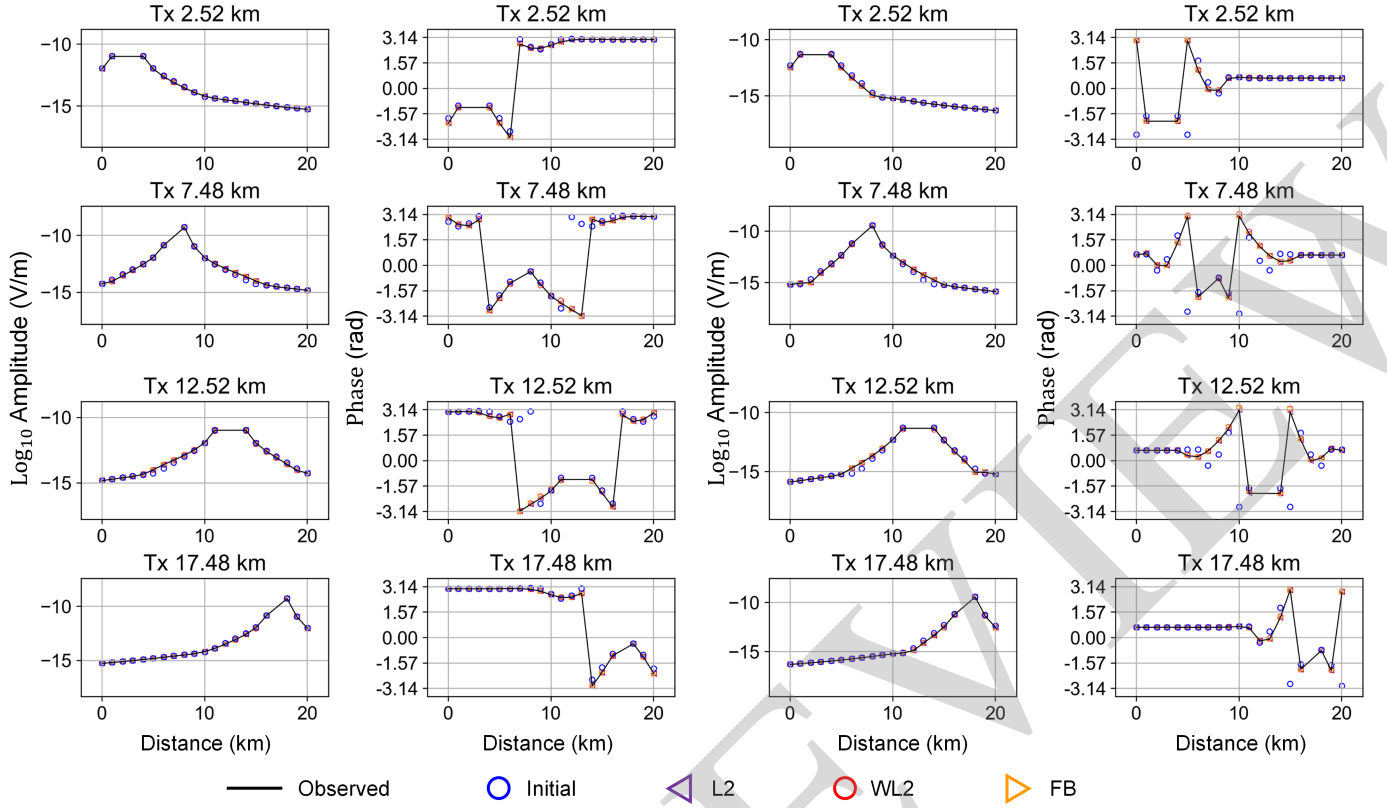


Fig. 8: The comparison between the observed data, initial data and data reconstructed using L_2 , weighted L_2 and feature-based inversion. For more detailed explanation, please see the main text.

while WL_2 inversion seems to be stuck in local minima after 35 iterations. In the feature-based inversion, the *projection steps* are iterations No. 2,4,6 and 8, and the pairs of points with the same horizontal coordinates on the FB curve in Fig. 7c respectively describe the data misfits before and after these projections.

Fig. 8 shows the comparison between the measured, initial and reconstructed data with different methods. In this figure, the four rows from top to bottom respectively correspond to transmitters at 2.52 km, 7.48 km, 12.52 km and 17.48 km, and the four columns from left to right respectively denote the data amplitude in logarithm for 0.25Hz, data phase for 0.25Hz, data amplitude in logarithm for 0.75Hz and data phase for 0.75Hz. The inverted data relatively satisfactorily reconstruct the observed data.

To find out how the reconstruction is updated in the feature-based inversion, we present a visualization of the evolution process in Fig. 9. The models updated with Gauss-Newton optimization in the i -th iteration are indexed $Ite-i$, and the projections of $Ite-i$ models to $\mathcal{R}(\mathcal{D})$ are indexed $Ite-i-Projected$. Comparing models before to after the projection, the non-uniform background as well as the conductive anomalies are filtered out and change to more uniform one, hence the non-uniqueness is alleviated and the local minima is avoided.

C. Experiment 2

We further conduct more experiments on other models to verify the effectiveness of the proposed inversion. The three

ground truth models are respectively shown in Fig. 10a-10c. In the former two models, the depth of the layers as well as the conductivity of the sea water, seabed and resistive layer are same with the Fig. 6a model. In Fig. 10a, the layer is slightly tilted, and in Fig. 10b the layer is divided into three sections. In the last model, the seabed conductivity is set to 0.7 S/m. The two resistive layers are both 4 km long, one is 120 m thick, located 800 m below the sea floor and has a conductivity of 0.05 S/m, while the other is 240 m thick, located 1500 m below the sea floor and has a conductivity of 0.01 S/m. The configurations of the DoI, mesh and transmitters and receivers are the same as above, and 5% noise is added to the simulated data in each case.

The reconstructed models by L_2 , WL_2 and FB inversion are respectively shown in the second, third and fourth rows in Fig. 10. The reconstructed traces and data misfit curves are displayed in Fig. 11. The conventional reconstructions basically exhibit similar patterns as shown above: unexpected conductive artifacts, unclear layer boundaries, inaccurate depth and exaggerated thickness. The advantages of feature-based inversion as exhibited above applies here as well. Besides them, we would like to highlight the generalization ability of the proposed method: the resistive layer with different shapes, such as inclined or composed of discontinuous segments, can be well delineated. We mainly attribute this to the fact that the adopted VAE can well balance the reconstruction ability and generalization ability. Pre-training enables the neural network to learn a wealth of prior knowledge on extracting image

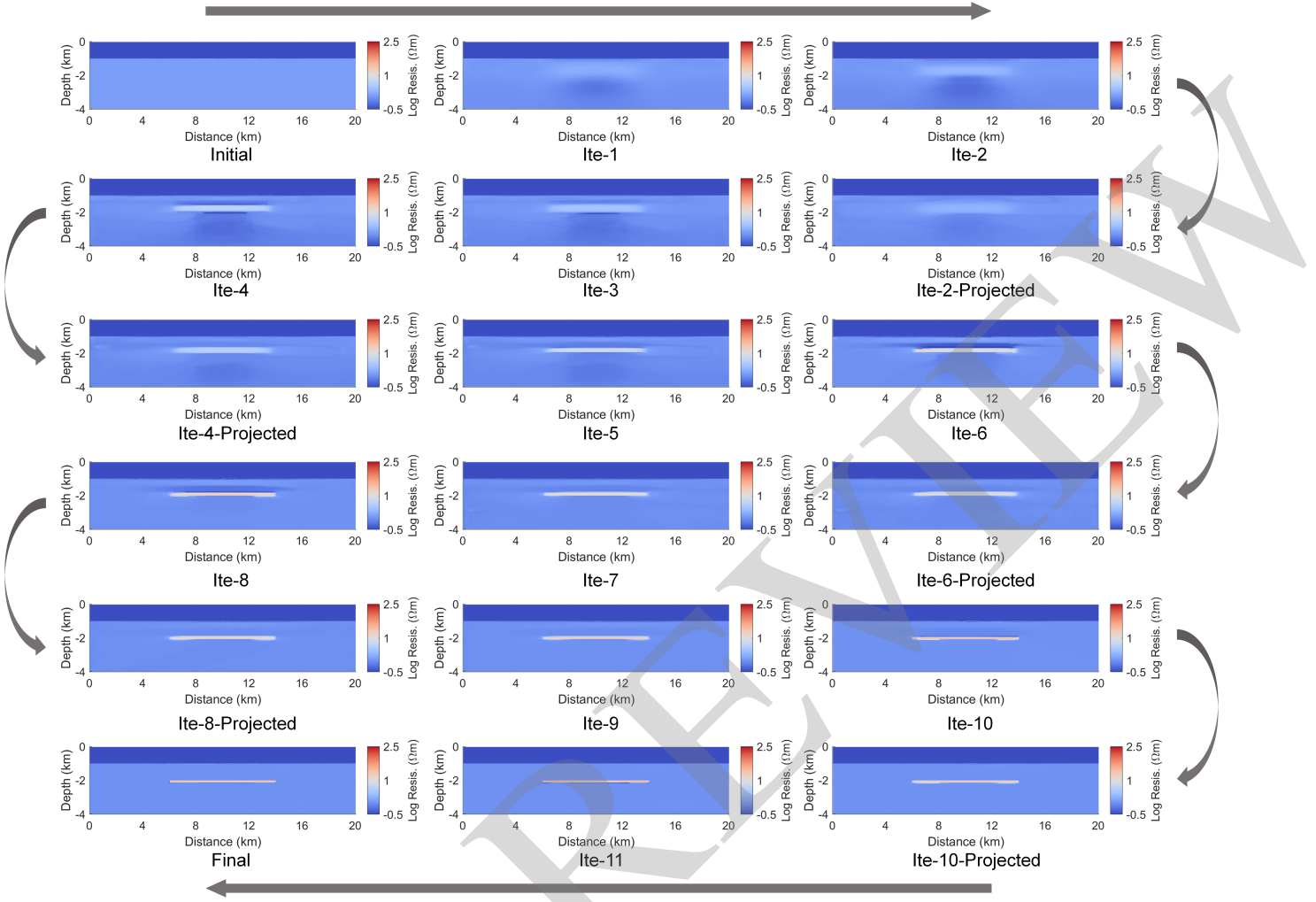


Fig. 9: The model evolution as shown in the experiment.

features and reconstructing images, while the very diverse natural images in the pre-training stage give the network strong generalization capabilities. In the fine-tuning stage, this prior knowledge is efficiently transferred to the distribution space of the conductivity model to be reconstructed.

Some metrics, including DM, IoU, MSE and SSIM as introduced earlier, are computed to assess the results and listed in Table I. *Model 1-4* respectively correspond to the four models, across which the L_2 and WL_2 reconstructions show limited difference in IoU. L_2 reconstructions presents better MSE than WL_2 , which may be explained by the observation that WL_2 inversion tends to introduce conductive anomaly with larger conductivities and thicker resistive layers compared to L_2 inversion. In contrast, WL_2 reconstructions outperform L_2 in SSIM, which may be explained by the observation that WL_2 yields a clearer boundary and more unique background compared to L_2 inversion. Both L_2 and WL_2 inversions have own features, however, in all the experiments, the FB reconstructions outperform them in all the three model domain metrics.

D. Experiment 3

In this experiment, we investigate the impact of the noise level on the feature-based inversion. The ground truth model is again Fig. 6a. The configurations of the DoI, mesh and transmitters and receivers are the same with above, but the noise level is changed to 15%, 25% and 50%, respectively. L_2 , WL_2 and FB reconstructions are shown in Fig. 12, and the reconstructed traces as well as the data misfit curves are displayed in Fig. 14. With the noise increasing from 5% to 25%, the L_2 reconstruction degrades: in Fig. 12b and 12c, the boundary of the resistive layer is still distinguishable, yet the background begins to be non-uniform. In Fig. 12d, the layer is hard to identify. For WL_2 inversion, with the noise increasing, the inversion is trapped in the local minima, causing the reconstructions convey limited information about the real conductivity distribution. For the feature-based inversion, higher noise lead to degradation in the reconstruction as well, e.g., the conductive artifacts appear (see the conductive jump in the FB curve in Fig. 13c and 13d), and the layer depth is depicted less accurately (see Fig. 13e and 13f the shift of the reconstructed resistive peak from the ground truth). However, even when the noise level is increased to 50%, the resistive

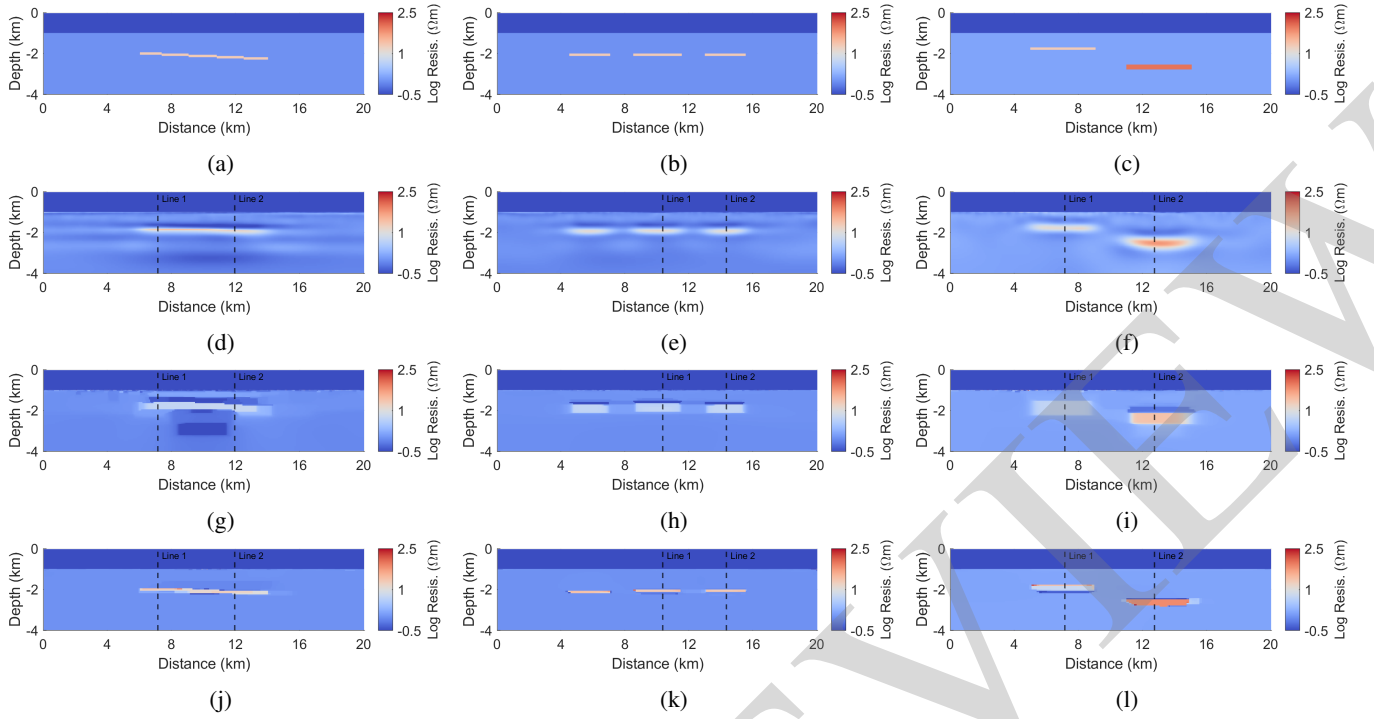


Fig. 10: The ground truth models and reconstructed models as introduced in Section III-C. The four rows from top to bottom respectively represent the ground truth models, reconstructed models with L_2 regularization, WL_2 regularization and feature-based inversion.

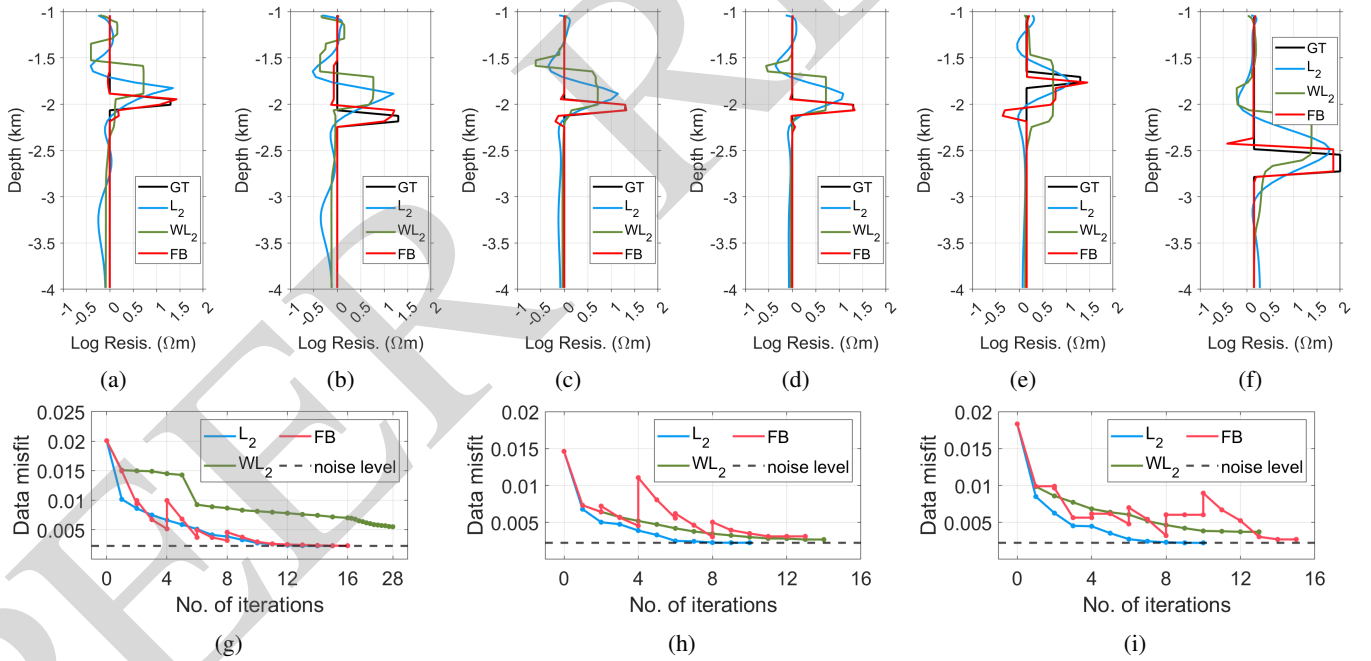


Fig. 11: 11a and 11b show the resistivity distributions in the two traces denoted with black dash lines as shown in Fig. 10a, and 11g display the data misfit during the reconstruction of 10a. Similarly, 11c, 11d and 11h show the reconstructed traces and data misfit corresponding to Fig. 10b model, and 11e, 11f and 11i show the reconstructed traces and data misfit corresponding to Fig. 10c model.

layer can be clear identified, which benefits the identification and interpretation of the reservoir using data with limited quality.

IV. FIELD EXPERIMENT

To further verify the studied inversion, we invert the CSEM data collected in the TrollWest Gas Province (TWGP) in

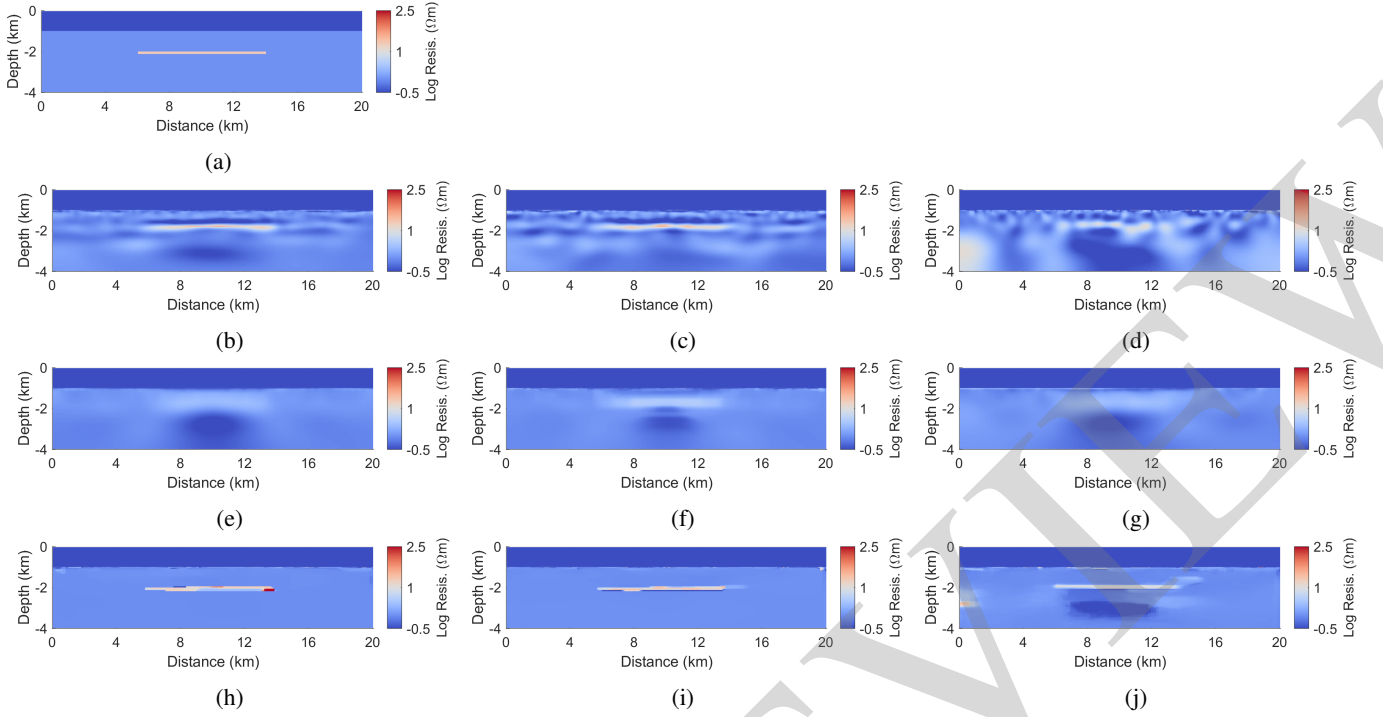


Fig. 12: 12a: Ground truth model. 12b-12j: the three rows from left to right represent inversion with L_2 regularization, WL_2 regularization and feature-based inversion, and the three columns from top to bottom represent the noise level of 15%, 25% and 50%.

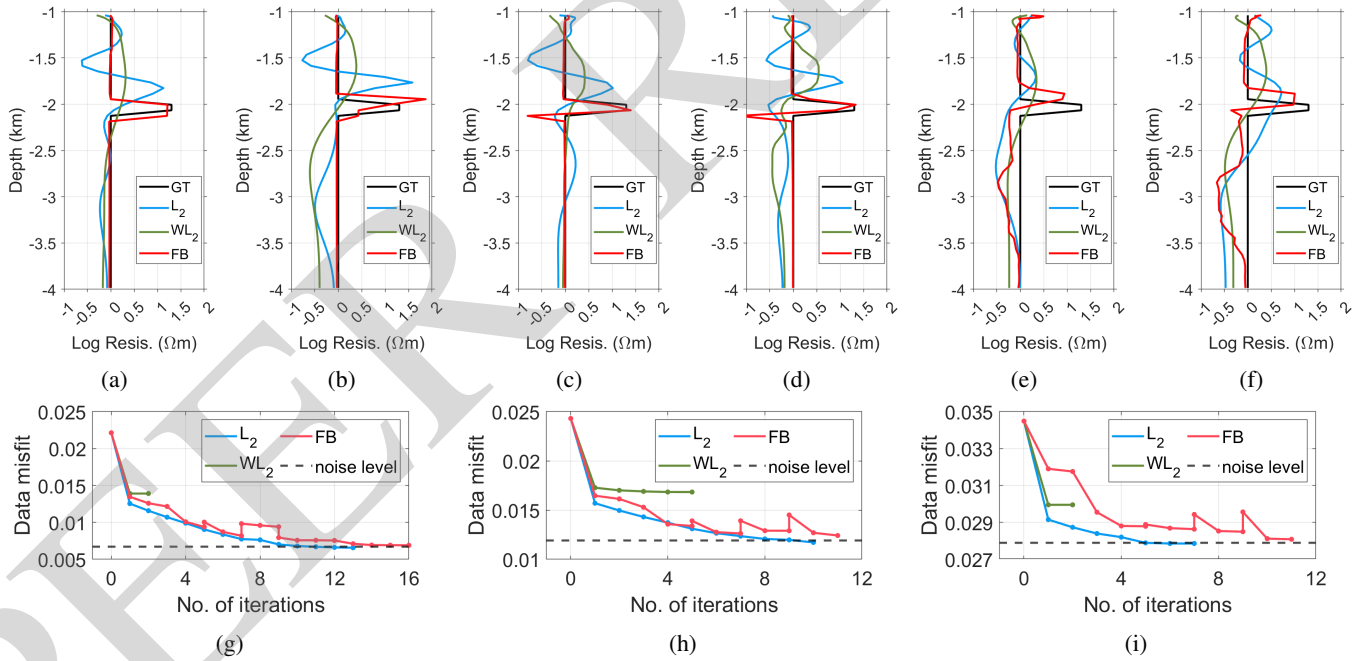


Fig. 13: Reconstructed traces and data misfit curves correspond to Fig. 12a model with data added noise of different levels. 13a, 13b and 13g correspond to 15% noise, 13c, 13d and 13h correspond to 25% noise, and 13e, 13f and 13i correspond to 50% noise.

Norway. This is a single-profile survey, and 24 EM receivers are distributed on the sea floor. The depth of the sea water is about 320 m. A 230 m long HED transmitter oriented along the survey line direction is towed by a ship and excite

EM field with 0.25 Hz continuous square wave signal at different locations, spanning a survey line of over 25 km. In this survey, gas reservoir is buried in Jurassic sandstone, with the maximum thickness about 160 m, a depth about

TABLE I: The comparison of data misfit (DM), intersection over union (IoU), mean square error (MSE) and structural similarity (SSIM) are shown in the table.

	Methods	DM	IoU	MSE	SSIM
Model 1	L_2	0.230%	0	0.0260	0.618
	WL_2	0.436%	0	0.0543	0.629
	FB	0.254%	0.339	0.0097	0.911
Model 2	L_2	0.227%	0.021	0.0234	0.619
	WL_2	0.551%	0	0.0432	0.628
	FB	0.230%	0.500	0.0079	0.907
Model 3	L_2	0.224%	0.146	0.0140	0.642
	WL_2	0.267%	0.200	0.0168	0.704
	FB	0.311%	0.675	0.0045	0.933
Model 4	L_2	0.222%	0.341	0.0191	0.840
	WL_2	0.372%	0.196	0.0523	0.853
	FB	0.269%	0.416	0.0110	0.957

1400 m and average resistivity around 200-500 Ωm [53]. The conductivity of water bearing Sognefjord Fm. sands and overburden sediments ranges from 0.5 to 2 S/m. More details about the survey can be found in [53], [54], and for existing inversion work on this survey, please refer to [2], [23].

In the inversion, we use EM data excited at 97 transmitter locations and measured at 23 receivers, and the distributions are plotted in Fig. 14a. Data at 0.25 Hz and 0.75 Hz is reliably extracted. The depth of the sea water is set to 320 m, and the DoI to be reconstructed ranges from 0.32 km to 3.5 km in depth, and from -7 km to 19 km horizontally. The model is discretized into 140 grids vertically and 261 grids horizontally, and the mesh used in the inversion is the same with forward modeling. The conductivity of the sea water varies from 4 S/m to 3.3 S/m [2]. Since its hard to accurately model the excitation strength, we invert the normalized logarithm amplitude (NLA) computed from data d with

$$\text{NLA}_{mnk} = \frac{\log_{10}|d_{mnk}|}{\sum_{m=1}^M \sum_{n=1}^N \log_{10}|d_{mnk}|}, \quad (27)$$

where m, n and k are respectively indices of transmitter, receiver and frequency. The NLA computed from the measurements at 0.25 Hz and 0.75 Hz are respectively shown in Fig. 16a and 16b. We also invert the phase in addition to NLA, which is not visualized due to the limited space. At 0.25 Hz, each receiver corresponds to 45 valid data on average, while at 0.75 Hz, each receiver corresponds to 23 valid data on average.

We present L_2 and FB reconstructions. A uniform 0.4 S/m model (see Fig. 14a) is used as an initial model for the L_2 inversion, and Fig. 14b shows the conductivity reconstruction in the L_2 inversion. The projected model of L_2 reconstruction to $\mathcal{R}(\mathcal{D})$ is shown in Fig. 14c and used as the initial model for FB inversion. The VAE parameters used here is the same with numerical experiments. Then, the feature-based inversion without extra model projections is conducted to reconstruct the conductivity (see Fig. 14d). For clarity, we select a vertical trace located at the offset of 7.45 km, and a horizontal trace located at the depth of 1.63 km (denoted by dashed lines in Fig. 14b and 14d), and plot the reconstructions in Fig. 15a and 15b. The curves of data misfit are presented in Fig. 15c. We can observe The overall conductivity pattern of TWGP can be

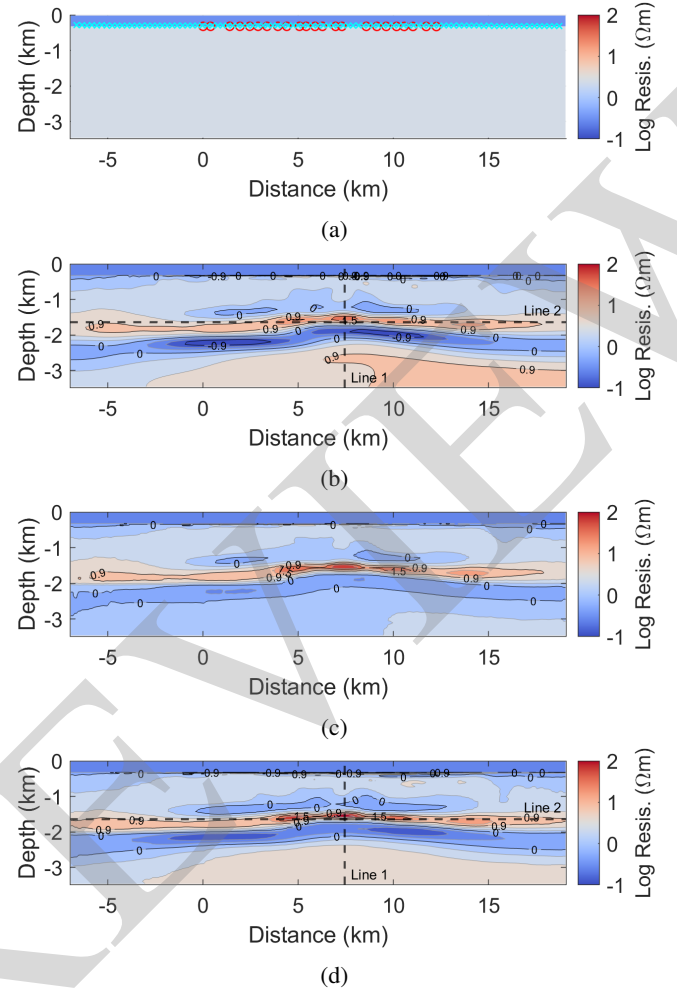


Fig. 14: Fig. 14a shows the initial model used in L_2 inversion. Fig. 14b shows the reconstructed model with L_2 inversion. Fig. 14c shows the initial model used in the feature-based inversion, which is the projection of Fig. 14b to the range of the VAE decoder. Fig. 14d shows the reconstructed model with feature-based inversion.

recovered by the conventional inversion, that is, the resistive layer buried in the conductive background, with the location and size basically consistent with the prior knowledge. However, the resistivity value deviates from the prior knowledge, affecting the delineation of the lateral size of the reservoir. Compared to conventional reconstructions, the features of the resistive layer are clearer in the projected model, which is well introduced into the FB reconstruction through the initial model. From FB reconstruction, we can find that the depth of the reservoir is about 1400 to 1600 meters, with the maximum thickness about 160 m, a typical thickness about 100 m, length about 8km and the maximum resistivity about 200 Ωm . The reconstructed patterns is basically consistent with prior knowledge, except that the maximum resistivity is slightly lower. The inverted NLA with conventional and proposed methods and corresponding data errors are displayed in Fig. 16. The final DM of the feature-based inversion is 2.37%, smaller than conventional inversion (3.38%).

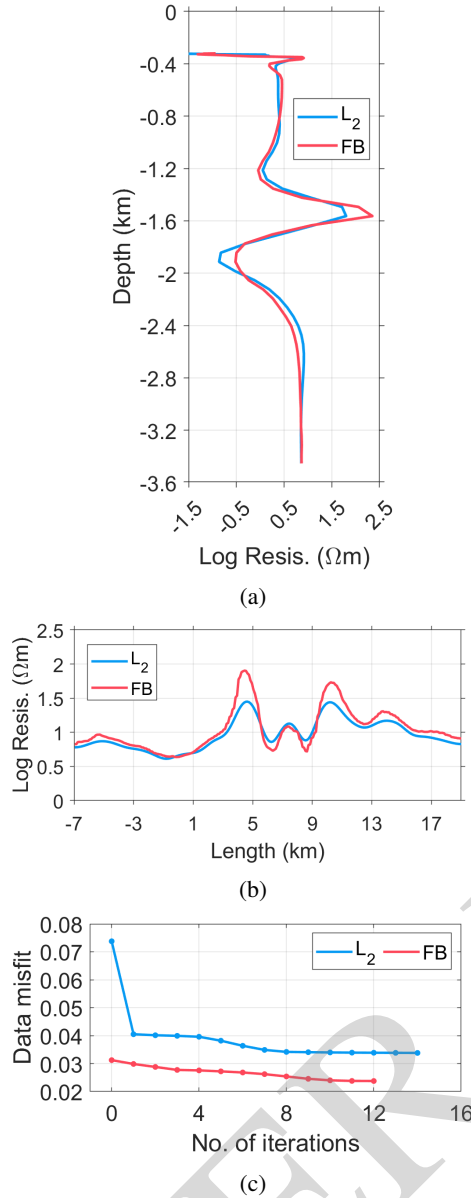


Fig. 15: Fig. 15a shows the vertical traces shown by Line1. Fig. 15b shows the horizontal traces shown by Line2. Fig. 15c shows the curves of data misfit in L_2 and feature-based inversions.

V. CONCLUSION

In this work, we study the feature-based 2.5D controlled source electromagnetic (CSEM) inversion using generative priors. Two-point-five dimensional modeling using finite difference method (FDM) is adopted to compute the response of horizontal electric dipole (HED) excitation in the 2D survey. Generative prior is incorporated in the inversion by reparametrizing the model with the variational autoencoder (VAE). In the inversion, the model is consecutively updated with Gauss-Newton method and projected to the range of the VAE decoder. Both numerical and field experiments verify that the proposed inversion is able to effectively incorporate

a priori knowledge and improve the reconstruction accuracy and possesses good generalization ability.

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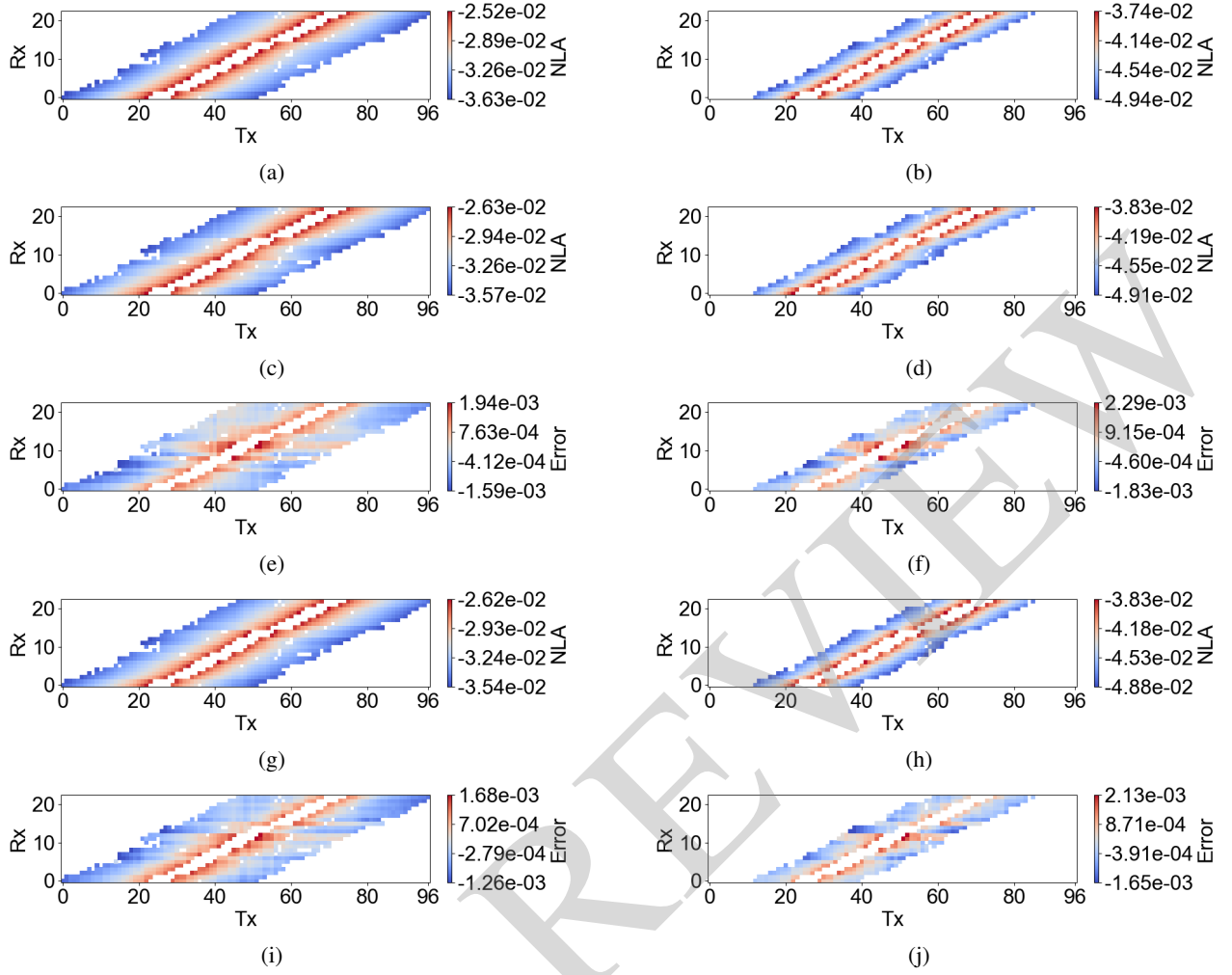


Fig. 16: Visualization of Normalized logarithm amplitude (NLA) data. The five rows respectively represent the measured data, data reconstructed with L_2 inversion, absolute error of L_2 inversion reconstructed data, data reconstructed with feature-based inversion, and absolute error of feature-based inversion reconstructed data. The two columns respectively represent 0.25 Hz and 0.75 Hz data.

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